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RESEARCH ARTICLE

Uncertainty Forecasting Model for Mountain Flood Based on Bayesian Deep Learning

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ABSTRACT Due to the characteristics of strong suddenness, high harmfulness, and frequent occurrence of mountain flood disasters in small watersheds, the accuracy and reliability of mountain flood forecasting are insufficient in small watersheds. This paper study key theories and technologies, that is the uncertainty forecasting model based on hydrologic physical mechanism. We design the Bayesian Deep Learning (DL) forecasting models, it is suitable for the transfer of spatiotemporal factors caused by mountain floods to disaster probability. The models include Bayesian Linear and Bayesian Long Short-Term Memory (LSTM) model, we hope to achieve an acceptable balance between reliability (uncertainty confidence coverage) and accuracy (confidence interval width). Meanwhile, we extract effective information from multi-source and multi-dimensional hazard factors' big data. The experiment shows the differences between Bayesian DL models, the models have long-term probability forecasting ability at both, but Bayesian LSTM is superior to Bayesian Linear in terms of reliability and accuracy.

INDEX TERMS Mountain flood forecasting, uncertainty forecasting, Bayesian deep learning, multi-source data, small watershed.

I. INTRODUCTION

Rainstorm and mountain torrents are sudden surface runoff caused by rainfall in mountainous areas. They are part of the most disastrous and difficult disasters for uncertain early warning, and also important factors affecting society and economy. According to the statistics of the World Meteorological Organization, the global casualty rate caused by rainstorm and mountain torrents has increased year by year, and the annual economic loss has reached hundreds of millions of dollars. 50.7% of China's land area is the rainstorm mountain torrent geological disaster prevention and control task area, in which about 570 million people are threatened by mountain torrents. Because of the characteristics of basin rainstorm and mountain torrents, such as strong suddenness, great harm and frequent occurrence [1]. For forecasting model of small watershed flood, physical models [2] and Artificial Intelligence (AI) models can be integrated to improve the physical process understanding and forecasting accuracy [3], [4], and adequate data analysis can promote the application of data [5], [6].

Hydrological and hydrodynamics' complex physical models are computationally demanding to calibrate the parameters [7], while AI models could enable fast parameter calibration [8], sensitivity analysis and interval estimation [9], [10], such as Machine Learning (ML), Deep Learning (DL) and Bayesian probability mode [11], [12], etc. AI models could help improve physical model outputs through

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model observation mapping and downscaling to finer resolutions [13]. The parameterization schemes in physical models could be replaced by AI models to accelerate computation with relatively simple model structure [14], [15]. AI includes various intelligent algorithms for machines to solve the problems of optimization [16], searching, processing [17], learning and so on [18]. ML and DL are two powerful AI techniques to perform forecasting tasks [19], [20]. The ML algorithms train a model automatically using given datasets and make forecasting for unknown system states [21].

This study aims to develop a mountain torrents forecasting method based on the state-of-the-art Bayesian, DL and data-mining, to achieve high forecasting reliability and accurate measurement simultaneously. This paper studies key theories and technologies, that is the uncertainty forecasting model based on hydrologic physical mechanism. We design the Bayesian Deep Learning (DL) forecasting models, it is suitable for the transfer of spatiotemporal factors caused by mountain floods to disaster probability. The models include Bayesian Linear and Bayesian Long Short-Term Memory (LSTM) model, we hope to achieve an acceptable balance between reliability (uncertainty confidence coverage) and accuracy (confidence interval width). Meanwhile, we extract effective information from multi-source and multi-dimensional hazard factors' big data.

About the paper's structure, section II gives the related works for uncertainty forecasting model for mountain flood, section III presents our probability space forecasting theory using big data and Bayesian DL method. Followed by Section IV which give the study data and data analysis. Section V presents our model's experiment and results. The discussions and conclusions of this study are presented in Section VI and VII separately. This study shows the differences between Bayesian DL models, the models have long-term probability forecasting ability at both, but Bayesian LSTM is superior to Bayesian Linear in terms of reliability, accuracy and computational consumption.

II. RELATED WORKS

The current state-of-the-art network architecture is the so-called ''Long-Short-Term-Memory'' (LSTM) [22], [23], which is initial application in water conservancy prediction [24], [25]. Through a specially designed architecture, the LSTM overcomes the problem of the traditional Recurrent Neural Networks (RNN) of learning long-term dependencies representing [26], [27]. Storage effects within hydrological catchments, which may play an important role for hydrological processes [28], [29]. Ensembles refer to the use of a number of models with different structures instead of a single model [30], [31]. The concept of ensemble forecasting has been widely used in numerical forecasting [32], [33], hydrological prediction and earth system simulations [34], [35]. In terms of an improved DL model [36], [37], an improved LSTM flood forecasting model [42], which uses daily flow and rainfall as input data, and is suitable for large river basins flood forecasting [38], [39]. The LSTM-based codec model for multi-step flood forecasting in advance [40], [41]. Generally, they are only adapted to specific environments and have certain limitations.

The data (aleatoric) and model (epistemic) uncertainties construct the predictive uncertainty [42], [43]. The epistemic uncertainty could be reduced with more data (knowledge) and the aleatoric uncertainty could be regarded as reducible with the increase of measurement accuracy [44]. The data and model uncertainties could be modeled jointly. Bayesian forecasting system offers an ideal theoretic framework for uncertainty probabilistic flood forecasting via the deterministic hydrologic model [45]. It provides suitable theoretical structure, validated models and reasonable analytic method, and can be developed into various Bayesian forecasting approaches [46].

In addition, in view of uncertain factors, the Bayesian flood forecasting method is an advanced and effective flood estimation method [47]. Some emerging Bayesian forecasting methods (such as integrated Bayesian forecasting, Bayesian multi-model combination, etc.) overcome the limitations of a single model or fixed model weights [48], which will effectively reduce the uncertainty of forecasting and improve the robustness of forecasting [49]. In statistical models, the data uncertainty is generally expressed as Gaussian distribution, and the model uncertainty is represented by interval estimation [50]. The predictive uncertainty from data and models could be measured by the law of uncertainty propagation [51]. In AI models, the data uncertainty is usually modeled by a specific probability distribution and the model uncertainty is estimated by Bayesian inference [52].

Integrated hydrological models and data-driven AI models are gradually being applied to flood forecasting. Liu et al. [53] propose the integrated model chain for flood risk predictionSzeląg et al. [54] use the ML method, and develop a likelihood function simulator, the optimal combination of model parameters was obtained at last. Kapoor et al. [55] propose a sub-model hybridization of the GR4J rainfall-runoff model with DL architectures, such as Convolutional Neural Networks (CNN) and LSTM networks. The hybrid model is superior to the basic conceptual model and classical Deep Neural Network (DNN) architecture. Overall, the combination of hydrological models with DL models, data-driven intelligent model groups, and other methods can produce better mountain flood prediction results. However, there is no comprehensive theoretical basis for the reasonable combination of the model, and further theoretical improvement is needed.

Recently, Bayesian with modern variational inference has received increasing attention, such as Stochastic Variational Inference (SVI) and Markov Chain Monte Carlo (MCMC) [56], [57] for probability space's reliability and accuracy. A gradient-based SVI algorithm used in the topic model to discover the latent semantic structure of texts. SVI is a state-of-the-art approach that is easier and more efficient to approximate posterior distribution for large numbers of parameters than the traditional [58], the SVI could be

applied effectively in standard RNN to model time sequences of music [59]. Other variants of the SVI were also proposed to solve the problems of natural language processing and computer vision [60], [61]. MCMC sampling is commonly applied to approximate posterior distributions in the Bayesian approach [62]. However, MCMC is intractable for deep neural networks as it is slow to converge and requires high computational costs to estimate large numbers of parameters [63]. Nonetheless, A relatively balanced result is often not obtained in solving the reliability and accuracy of probability forecasting.

Finally, the accuracy of data and models is crucial. Szeląg et al. [64] use a limited range of input data for predicting sewer network flooding and modernization with the DNN model, and obtained good results. Pingping Luo, et al. [65] propose that accuracy and computational efficiency are two key areas, that hinder the improvement of urban flood model quality. Jahangir et al. [66] propose the model performance sensitive to precipitation forecasting accuracy. DL models 's accuracy can be improved by optimization the input data and hyperparameter [67]. In short, when designing a prediction model, it is important to consider input and parameter tuning to improve the accuracy of the forecasting.

III. METHODOLOGY

A. THEORY

Frist, a spatiotemporal factor structure affecting the occurrence of mountain flood disasters in small watersheds is constructed, based on hydrological forecasting theory. Second, a qualitative relationship between the main factors and the time delay effect of flood water levels is established, and a mathematical model for mountain flood disasters in small watersheds is constructed, by analyzing the historical data of various disaster causing factors and water levels in typical small watersheds. Finally, we select the basic mathematical model, based on maximum approximation and variational method to achieve uncertainty transfer forecasting mechanism, and derive it from the Bayesian DL algorithm model to establish a DL forecasting mathematical model.

Vrugt and Beven [68] proposed a process-based hydrological model to produce streamflow in the outlet of the catchment can be expressed as:

$$y_0 = y_m\left(\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{\alpha}}, \tilde{\boldsymbol{\Psi}}, \tilde{\boldsymbol{\omega}}_0; \tilde{\boldsymbol{\theta}}, \tilde{\boldsymbol{\phi}}\right) + y_\varepsilon \tag{1}$$

$\tilde{\boldsymbol{X}}, \tilde{\boldsymbol{\alpha}}, \tilde{\boldsymbol{\Psi}}, \tilde{\boldsymbol{\omega}}_0, \tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\phi}}$ are the spatiotemporal data of hydrological and other disaster causing factors, constant variable vectors, spatial measurement values, initial state vectors, general parameter vectors, and spatial parameter arrays. As errors, represent the time series vectors of water flow observed through hydrological models.

Based on the dynamic characteristics of hydrological data, derive a basic residual neural network DL model:

$$y_\varepsilon = f(\boldsymbol{X}, \boldsymbol{\theta}) + \varepsilon \tag{2}$$

and consider the Bayesian posterior probability factor with time delay:

$$p(\boldsymbol{\theta} \mid y_0) = \frac{p(y_0 \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{p(y_0)} \propto p(y_0 \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}) \tag{3}$$

Using Kullback–Leibler (***KL***) divergence to measure the distance between these two distributions, achieving approximate variation, i.e. optimization:

$$\boldsymbol{KL(q||p} = \sum_{i=1}^{n} \boldsymbol{q(\theta) log} \frac{\boldsymbol{q(\theta)}}{\boldsymbol{p(\theta|X, Y)}} \tag{4}$$

Variational estimation based on (4), dynamically adjusting ***X, Y***, and parameters θ, Improving the forecasting performance of big data and AI, that is, dynamically adjusting ***X*** and ***Y*** through the effective information extraction technology proposed by the applicant for new multi-source and multi-dimensional uncertainty forecasting factors, and dynamically adjusting parameters through the construction technology of new DL forecasting models $\boldsymbol{\theta}$, it be expected to significantly improve predictive performance.

B. METHOD

We construct a deterministic and uncertain composite forecasting model based on multidimensional water conservancy data. We designed a DL framework based on multiple technologies such as Linear and LSTM to train and continuously correct composite models, a deterministic DL model scheme for each forecasting point is constructed. The dynamic Bayesian method is used to construct an uncertainty probability DL forecasting supplementary model based on historical prior information and parameter periods, as shown in Figure 1.

At the multi-source data block, the historical data time series set and real-time data stream are selected, divided into training set and test set, and the real-time data stream is transformed to forecasting's input data. The input data need standardized processing, data segmentation for DL model and Bayesian block for uncertainty forecasting. DL block include the model training and continuous correction. Among them, X is the input; cells are including the hidden multi-layer DL units; P is the actual output; C, h are the input and forget gate parameters; L is the number of cells. The training process include the theoretical output, loss calculation and actual output. The model achieves optimal parameters for each layer through repeated training, validation, and the use of Adam error training method. Bayesian block's main function of is training for Bayesian parameter estimation, including weight estimation (W) and Bias estimation (b). Finally, the result block: Water level/hazard level prediction and measurement time series set, a forecasting model library is constructed to predict multiple small watersheds for Short-Term (0∼24 hours), Mid-Term (1∼7 days) and Long-Tern (1∼30 days)'s eater level and disaster level.

The general Liner method is structed by multi-layer perceptron. For a one-hidden-layer Liner whose hidden layer has hidden units, we denote by the outputs of the hidden layer,

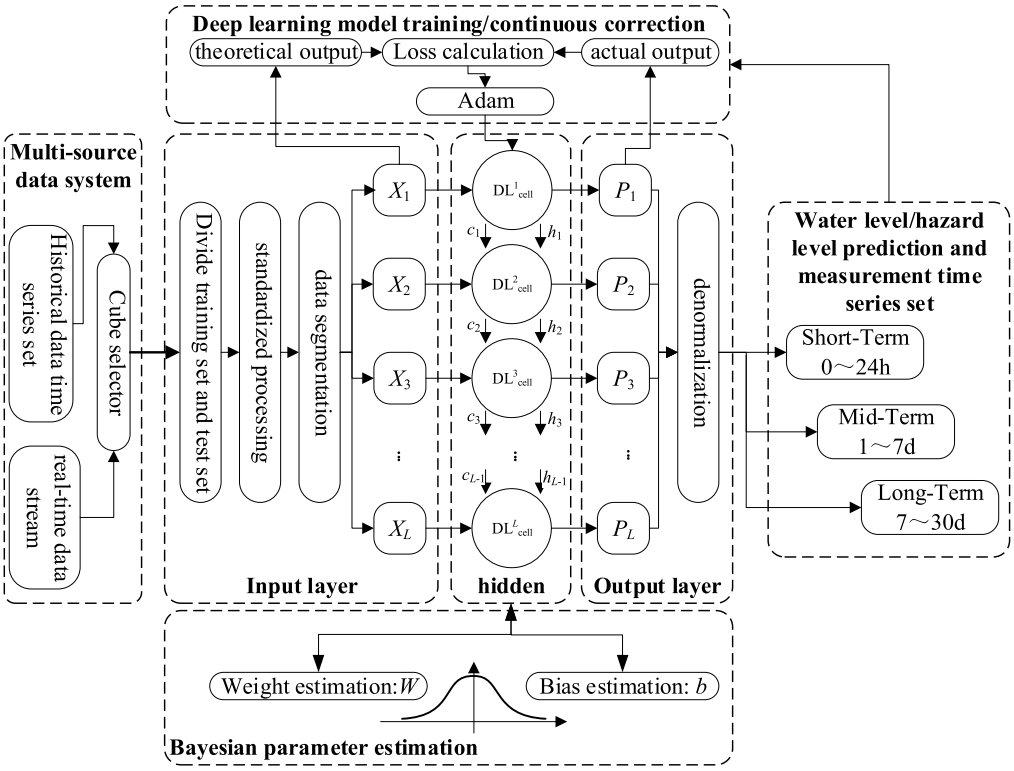


FIGURE 1. The architecture of Bayesian DL model.

which are hidden representations. Since the hidden and output layers are both fully connected, we have hidden-layer weights and biases and output-layer weights and biases. This allows us to calculate the outputs of the Liner as follows, and the parameter's interpretation is shown in Table 1.

$$h_t = \sigma(W_i x_t + b_i) \tag{5}$$

$$O_t = W_O h_t + b_O \tag{6}$$

The LSTM method can preserve information over long periods through its unique structure, cell state, and gates to control how the information flows in the different layers. The structure of the LSTM is shown in Figure 2.

Equations are as follows:

$$i_t = \sigma\left(W_i x_t + U_i h_{t-1} + b_i\right) \tag{7}$$

$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{8}$$

$$o_t = \sigma\left(W_O x_t + U_O h_{t-1} + b_O\right) \tag{9}$$

$$\tilde{C}_t = \tanh\left(W_c x_t + U_c h_{t-1} + b_c\right) \tag{10}$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tilde{C}_t \tag{11}$$

$$h_t = O_t \otimes \tanh\left(C_t\right) \tag{12}$$

The parameter's interpretation is shown in Table 1.

The input gate filter is the information added to the cell with the assistance of current input and historical information, while the forget gate discards certain previous information, which allows the cell to clear the value it contains; subsequently, the output gate outputs the information selectively.

TABLE 1. Parameter's interpretation.

Param	Interpretation
x_t	The input data, including hydrological, meteorological, geographic, and crowdsourced data on the same time scale.
i_t	The input gate
f_t	The forget gate
o_t	The output gate
W_i	The weight of connecting input data and the input gate
W_f	The weight of connecting input data and the forget gate
W_o	The weight of connecting input data and the output gate
U_i	The weight of connecting input gate and the hidden state
U_f	The weight of connecting forget gate and the hidden state
U_o	The weight of connecting output gate and the hidden state
b_i	The input gate bias vector
b_f	The Forget gate bias vector
b_o	The output gate bias vector
C_{t-1}	The cell input
C_t	The current cell state
h_t	The current hidden state
σ	The logistic sigmoidal function
$\otimes$	The element-wise multiplication (dot product)
tanh	The hyperbolic tangent function

As we know, on deterministic (non-Bayesian) neural network layers, the trainable parameters correspond directly to the weights used on its linear transformation of the previous

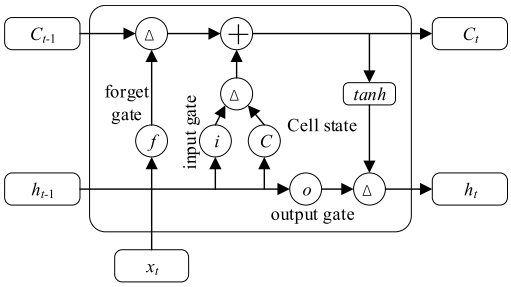


FIGURE 2. The architecture of the LSTM model.

one (or the input, if it is the case). It corresponds to the following equation (13), Z correspond to the activated-output of the layer i.

$$a^{i+1} = W^{i+1} * z^i + b^{i+1} \tag{13}$$

According to formula (4), adjust the parameters $\boldsymbol{\theta} = \{W, b\}$, then better forecasting can be obtained. $\boldsymbol{X}$ is the input based on (5); cell is the hidden multi-layer DL unit; P is the actual output; c, h are the input and forget gate parameters; L is the number of cells. For the parameter $\boldsymbol{\theta}$ in equation (3), take the approximate variation of Gaussian distribution for equation (4), then the weight and bias distribution on the i-th layer of the model for the times are:

$$W_n^i = \mathcal{N}(0, 1) * \log\left(1 + \rho_W^i\right) + \mu_W^i \tag{14}$$

$$b_n^i = \mathcal{N}(0, 1) * \log\left(1 + \rho_b^i\right) + \mu_b^i \tag{15}$$

where the sampled W and b corresponds to the weights used on the linear transformation for the i-th layer on the i-th sample. ρ parametrizes the standard deviation and μ parametrizes the mean for the sample linear transformation parameters.

Bayesian layers seek to introduce uncertainty on its weights by sampling them from a distribution parametrized by trainable variables on each feedforward operation. This allows we not just to optimize the performance metrics of the model, but also gather the uncertainty of the network forecasting over a specific datapoint (by sampling it much times and measuring the dispersion) and aiming reduce as much as possible the variance of the network over the forecasting, making possible to know how much of uncertainty we still have over the label if we try to model it in function of our specific datapoint.

It is possible to optimize our trainable weights. Even though we have a random multiplier for our weights and biases, it is possible to optimize them by, given some differentiable function of the weights sampled and trainable parameters (loss), summing the derivative of the function relative to both of them:

$$\Delta_\mu = \frac{\delta f(\omega, \theta)}{\delta \omega} + \frac{\delta f(\omega, \theta)}{\delta \mu} \tag{16}$$

$$\Delta_\rho = \frac{\delta f(\omega, \theta)}{\delta \omega} \frac{\epsilon}{1 + e^\rho} + \frac{\delta f(\omega, \theta)}{\delta \rho} \tag{17}$$

$\epsilon = \mathcal{N}(0, 1), \theta = (\rho, \mu), \omega = \mu + \log(1 + e^\rho) * \epsilon, f(\omega, \theta)$ be differentiable relative to its variables.

It is also true that there is complexity cost function differentiable along its variables. It is known that the cross-entropy loss (and MSE) is differentiable. Therefore, if we prove that there is a complexity-cost function that is differentiable, we can leave it to our framework take the derivatives and compute the gradients on the optimization step.

The complexity cost is calculated, on the feedforward operation, by each of the Bayesian Layers, (with the layers pre-defined-simpler priori distribution and its empirical distribution). The sum of the complexity cost of each layer is summed to the loss.

we can gather the complexity cost of a distribution by taking the ***KL*** divergence from it to a much simpler distribution, and by making some approximation, we will can differentiate this function relative to its variables (the distributions).

IV. DATA

A. STUDY AREA'S DATA

The Tiaoxi River originates from the mountainous city of Hangzhou, Zhejiang province, with a main stream length of 157 kilometers. The rainy season lasts from the middle of May to the middle of July every year, with a precipitation of 450-510mm; August to September is the flood season. The rainfall is 190-380mm, and the maximum 24-hour point rainfall is 681.2mm (Shiling station, August 1, 1956). The rainfall in the flood season accounts for about 75% of the annual rainfall. The heavy rain in East and West Tiaoxi basically occurs synchronously, and the rainfall is also relatively close. This paper choice Hourly data for 3 years of typical mountain flood monitoring point, including weather, rainfall, temperature, humidity, wind direction, wind speed and other meteorological data; rainfall, water level and other hydrological data; weather forecast data. The hydrological data mainly come from the water conservancy management department of Zhejiang Province (https://sqfb.zjsq.net.cn). Daily streamflow and meteorological data (including precipitation, mean air temperature, maximum air temperature, minimum air temperature, wind speed, relative humidity, and solar radiation) are collected from the hydrological station and the China Meteorological Data Service Centre (https://data.cma.cn), respectively. We also collect the geography and human activities data, even if their influence is small.

B. DATA PREPROCESSING

We connect to the multi-source data of the main influencing factor data platform such as hydrology, meteorology, geography, human activities, and combine with the R&D team's water conservancy database to build a data center to achieve large-scale collection of historical and real-time data. After data preprocessing, data fusion, storage and other processes for the compound forecasting model to use, and model parameters into the model database, the overall plan of multi-source data processing is shown in Figure 3.

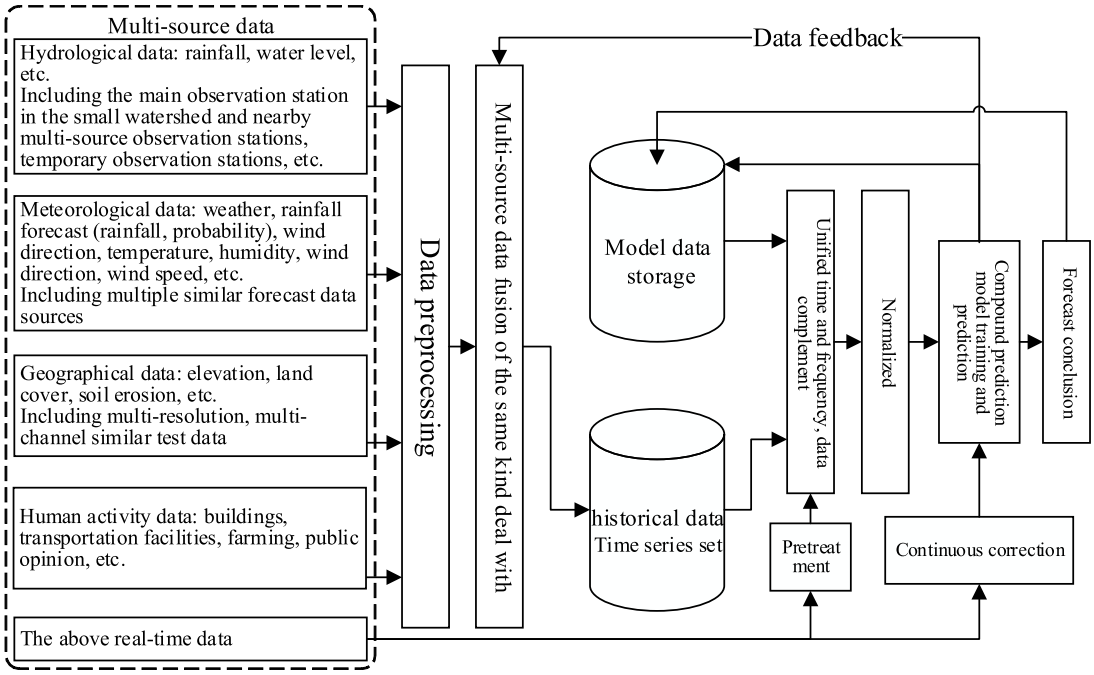


FIGURE 3. Overall scheme of multi-source heterogeneous data processing.

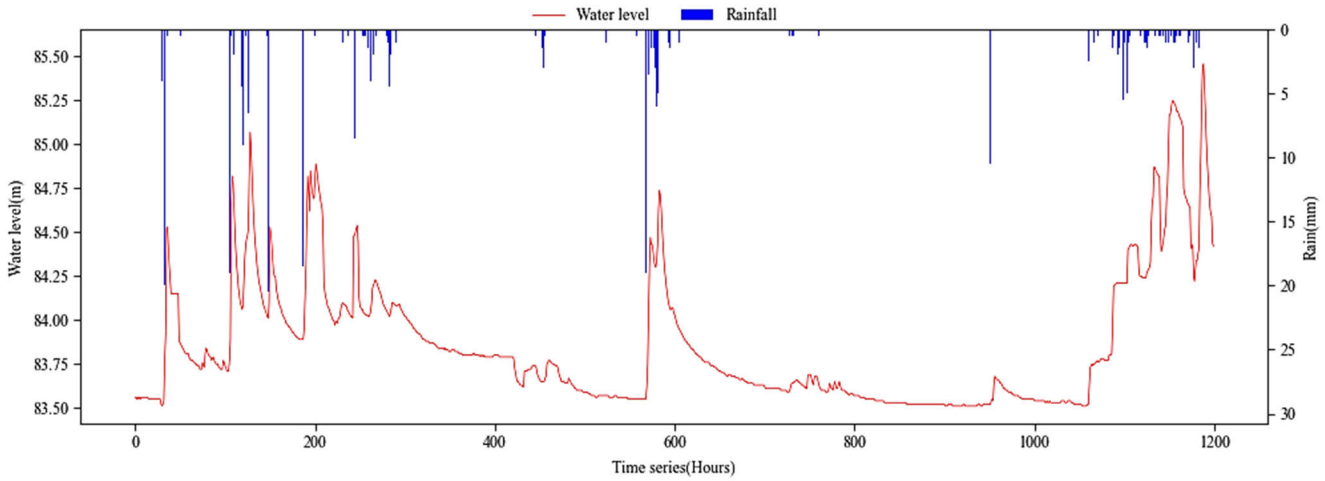


FIGURE 4. Study area's main data (water level and rainfall).

In addition, large-scale instruments such as terrain scanning drone terrain collectors, multi-wave speed river terrain survey boats and other large instruments are used to collect high-density geographic data in the test area; a water conservancy data mobile Internet of Things device is built at the test site to collect real-time hydrometeorological and geographic data to improve the real-time and accuracy of model input data. We chose data from 1250 consecutive hours to cover over the flood period about data from June 9th to July 31th, 2022, with the main rainfall and water levels shown in the Figure 4.

It is obvious that the periodicity and regularity of water level changes are not obvious, and conventional regression data forecasting methods are not feasible. It is necessary to fully explore input information. Overall, the input data has the following characteristics:

(1) There is a great deal of uncertainty in the disaster causing factors, and it is necessary to ensure the reliability of the data through data filling, mutual verification, and peer verification methods.

(2) Rainfall data and water level data follow a normal distribution, and the water level has a certain periodicity, which is suitable for Bayesian probability forecasting.

(3) Although the data interval is 1 hour, the smoothness of predicting data changes is smooth on a large scale, but the high discreteness and fast change on a small scale

TABLE 2. The models' training parameters.

Parameter	Value	Meaning
units	64	Positive integer, dimensionality of the output space.
Learning Rate	Adam (Adaptive Learning Rate)	The proportion of inputs that need to be discarded.
activation	relu	Activation function
kernel_regularizer	tf.keras.regularizers.l2()	Regularizer function applied to the kernel weights matrix.
adam	0.0001	Adam optimization function and value.
loss	mean_squared_error	Loss function
metrics	accuracy	List of metrics to be evaluated by the model during training and testing.
batch_size	8	Integer or None. Number of samples per batch of computation.
epochs	500	Number of epochs to train the model.

may affect the continuity of the forecasting probability sharpness.

The input variables were scaled to accelerate the convergence speed and make the training process faster and were normalized using the following equation:

$$\hat{\boldsymbol{x}} = \frac{\boldsymbol{x} - \boldsymbol{\mu}}{\boldsymbol{\delta}} \tag{18}$$

where $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$ are the observed and z-score standardization data of water-levels and relation features at time t, respectively.

V. EXPERIMENT AND RESULTS

A. EXPERIMENT ENVIRONMENT

Python is a programming language that can provide DL models, and with Tensorflow 2.0 and PyTorch library are efficient tools for predictive data analysis. The extensible library to create Bayesian Neural Network layers on PyTorch at GitHub (https://github.com/) is used. Calculations were carried out with the CPU of intel(R) Xeon(R) E-2176M, double cores,2.70GHz and 32 GB RAM.

The ensemble DL model include Bayesian Linear (BR) and Bayesian LSTM (BL), and the two Bayesian DL model have the similar structures, the only difference is the type of data DL layer, one is a linear mechanism and the other is an LSTM structure. The models have the main parameters' value and meaning at Table 2. The parameters are seted as the best result obtained through adaptive calculation based on the model Additionally, it should be noted that modeling boundary conditions are kept as similar as possible to allow comparison of different models. This means that input data, number of layers, number of neurons, dropout rate, and number of training epochs are kept identical. Therefore, each model acquired a different set of parameters with the same model structure, which show the information from different catchments that could be analyzed based on simulation results. After training, the value is denormalized for comparison with observed data.

In this section, the comparative performance analysis of Bayesian methods for long-term forecasting is presented. Compared with the uncertainty models using various statistics measure. Confidence Interval Level (CIL), it is the ratio of the point where the actual value falls within the confidence interval to the actual value, which means the interval coverage probability of Bayesian DL's output. The CIL is higher, the higher the reliability of the prediction. and Confidence Interval Average Width (CIAW), which means the sharpness of the Bayesian DL's output. CIAW is the average width of a specific forecasting interval, the smaller its value, the higher the accuracy.

We use Bayesian Linear and Bayesian LSTM for comparative analysis, mainly focusing on CIL (i.e. reliability) and CIAW. However, CIL and CIAW are a pair of contradictory results:

(1) We increase the width of the confidence interval improves reliability and reduces accuracy.

(2) We reduce the confidence interval width reduces reliability and improves accuracy.

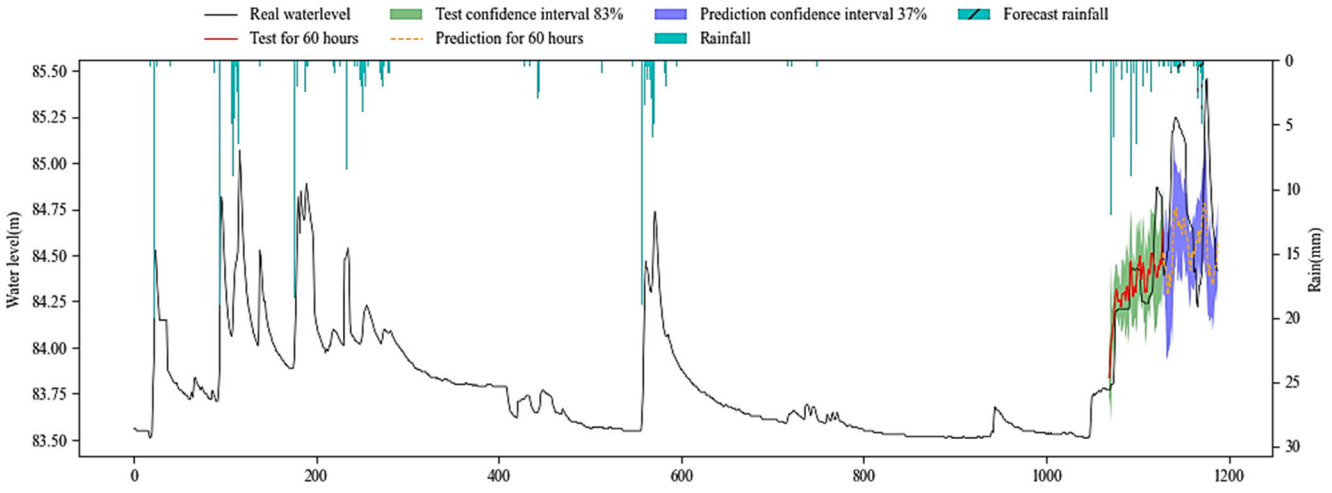
The experiment includes the training, test and prediction phase, they respectively account for 80%, 10%, and 10% of the entire time series data, we gradually improve the effectiveness of predictions through these three-tiered stages. We hope to achieve an acceptable balance between CIL and CIAW, the larger the probability value of its confidence interval, the more reliable it is, and the narrower the confidence width, the more accurate it is.

B. RESULTS

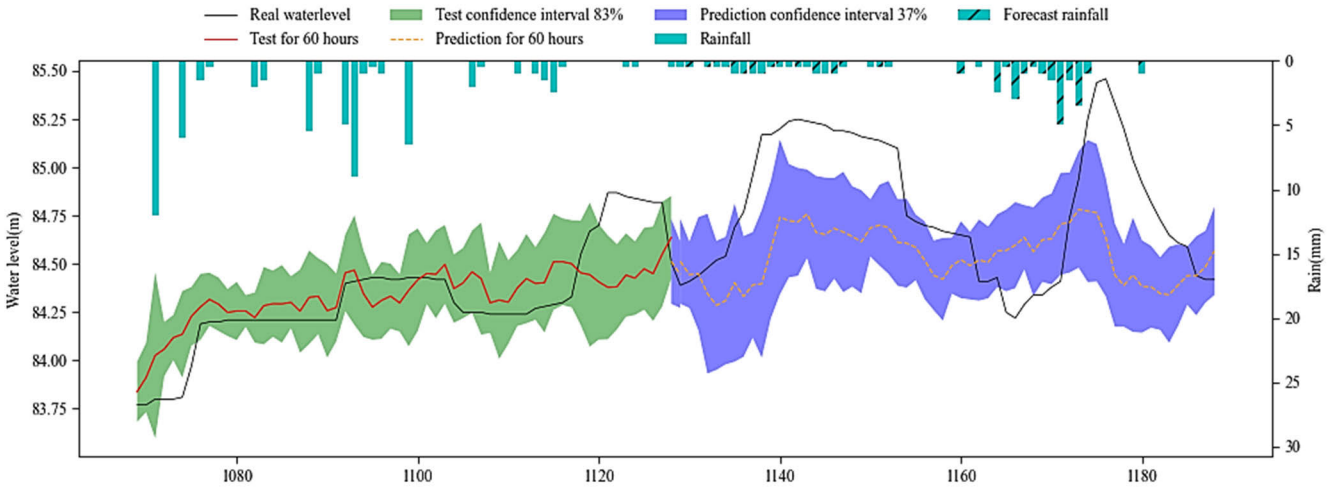
Figure 5 and Figure 6 show the overall effect of Bayesian Linear and Bayesian LSTM, as well as the details of the prediction stage, respectively, the Figure (a) show the Training, test and prediction phases' Overall graph, the Figure (b) show the test and prediction phases' Detail graph. Table 3 explained the meaning, range, and color of each data of the label in the figures.

Although rainfall is the main factor affecting water levels, water levels and rainfall data do not exhibit significant periodic variability, we must use 80% of the sample data including hydrology, meteorology, geography and human activities data, for training in order to improve the accuracy of prediction.

It can be observed from the training, test and prediction phases' overall graph that although the mountain flood water level changes rapidly, the overall trend of the two uncertainty forecasting methods is basically consistent with the actual water level, indicating that the uncertain forecasting model for mountain flood based on Bayesian DL is feasible. The error in the later stage of BR gradually increased, resulting in a serious dislocation between the confidence coverage interval and the actual value in the prediction stage, while the



(a) Training, test and prediction phases' overall graph



(b) Test and prediction phases' detail graph

FIGURE 5. Bayesian linear model's training, test and prediction phases.

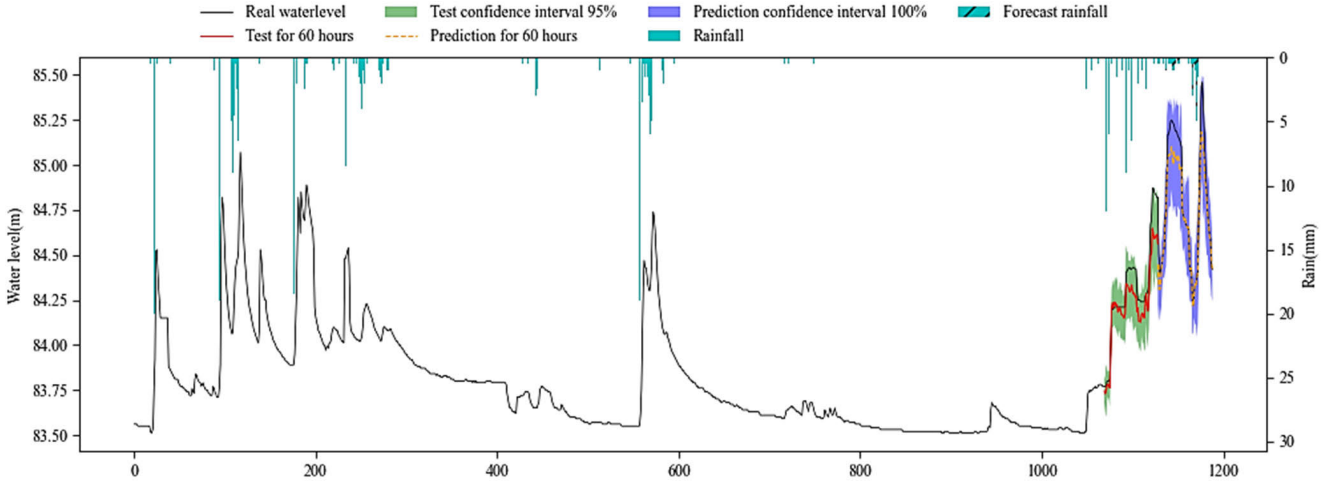
TABLE 3. Forecasting accuracy and durations of improved models.

Data	Meaning	Range	Position / Color
Water level (m)	Water level Based on the elevation of 85 Yellow Sea	83~85.5	y-axis
Time series (Hours)	Data from June 10th to July 30th, 2022	0~1200	x-axis
Rainfall	Hourly rainfall (mm)	0~25	green
Real waterlevel	Data for training of water level	83~85.5	Black
Test for 60 hours	Confidence interval median for testing of water level	83~85.5	Red
Test confidence interval	Confidence interval for prediction Within 60 hours	0~1	Green
Forecast rainfall	The rain of predict by Meteorological department	0~25	Blue
Prediction confidence interval	Confidence interval for prediction Within 60 hours	0~1	purple
Prediction for 60 hours	Confidence interval median for prediction of water level	83~85.5	Orange

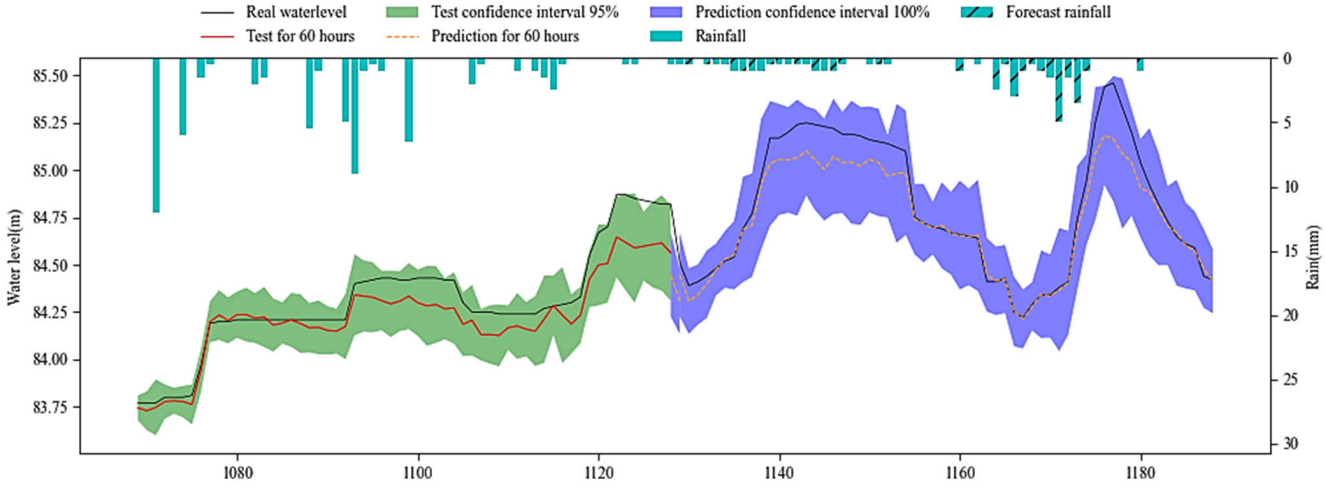
CIL and CIAW of BL were relatively stable in both the test stage and the prediction stage, the confidence interval median for testing of water level and Confidence interval median for prediction of water level have the same change rule with the CIAW.

In terms of specific data comparison, Table 4 shows the differences between these two algorithms. BL is superior to BR in terms of CIL, CIAW, and computational consumption. Specifically, in terms of CIL, BL during training is 95%, while BR can only reach 83.33%, and BL can still maintain 100% reliability during prediction. In terms of CIAW, BL's CIAW is smaller than the BR. BL's accuracy is better than BR. in both training and prediction stage. The key reason is that the effectiveness is insufficient without sufficient information mining of the BR.

From the perspective of time, the training time of the two prediction models is 53 and 50 minutes, which is less than 1 hour, and can adapt to the frequency of mountain



(a) Training, test and prediction phases' overall graph



(b) Test and prediction phases' detail graph

FIGURE 6. Bayesian LSTM model's training, test and prediction phases.

TABLE 4. Comparison of effects of bayesian DL model.

No	Type	Meaning	Bayesian Linear (BR)	Bayesian LSTM (BL)
1	Test's CIL	Test's Confidence Interval Level, it is proportional to the reliability of the results.	83.33%	100.00%
2	Test's CIAW	Test's Confidence Interval Average Width, it is inversely proportional to the accuracy of the results.	0.42m	0.32m
3	Prediction's CIL	Prediction's Confidence Interval Level, it is proportional to the reliability of the results.	36.67%	95.00%
4	Prediction's CIAW	Prediction's Confidence Interval Average Width, it is inversely proportional to the accuracy of the results.	0.50m	0.4764m
5	Computing time	Computing time for training.	53min	50min
6	Steps of training	Computing steps of training when the error is stable.	401	123

torrents prediction every hour. The steps of BL training are only 123 times, which is less than 401 times of BR, that is, the training error of BL can converge quickly and reach a stable value, and BL model has certain advantages in time consumption.

VI. DISCUSSIONS

Generally, as the forecasting time prolongs, the accumulation of errors in Bayesian DL mode forecasting increases, and the accuracy of forecasting deteriorates. But Bayesian LSTM has achieved highlighted results. The main reason for this effect

is the loop extraction information of LSTM, which reduces errors and improves forecasting accuracy. In addition, the computational complexity and time of the entire dynamic process are acceptable, and can be concurrently applied to thousands of small watershed water level forecasting points on a large scale to obtain cross forecasting and more accurate results. According to the experimental analysis, it can be seen from the figure that the comprehensive effect of the whole confidence interval is guaranteed, but the peak forecasting effect is not ideal, which is a very important index for water level forecasting.

In order to illustrate the complexity of changes in mountain flood water level, compared with literatures [68], [69] and [70], the data collection period is based on year or month, and they are very regular, have low perplexity, and relatively low forecasting difficulty. So that, they are not suitable for predicting mountain flood water levels. Of course, the Bayesian DL proposed in this article is not flawless. In order to maintain the accuracy of the model, data preprocessing is necessary, and the model needs to be continuously trained and calibrated. Otherwise, the error accumulation will be serious, which will gradually reduce the reliability and accuracy of forecasting.

In addition, in the case of complex changes in water level, many literatures show relatively periodic changes in data on a large scale, such as year, month, week and day. Although there is similarity between years, the actual water level is low periodic, which is also in line with the actual situation. This type of forecasting has a certain level of complexity and is often more difficult than stock data prediction, sometimes approaching noise. Moreover, with the extension of forecast time, more and more errors accumulate, the reliability and accuracy of forecast becomes worse and worse.

Comparison with input data [64], neglect of soil moisture can lead to errors in estimates of flood volume. rainfall depth, rainfall duration are the most important rainfall data for the prediction of sewer flooding, adding some data can still increase the reliability and accuracy of the data.

Reliability and accuracy are a pair of contradictions, which improve the reliability of mountain flood forecasting, but may sacrifice accuracy and fail to achieve the purpose of forecasting. If accuracy is simply improved while reliability is reduced, this will not be adopted in the water conservancy industry. Water conservancy disaster prevention and reduction are mainly aimed at ensuring social safety.

Another limitation of this article is that although it has been estimated based on hydrological models, which are more interpretable than general data-driven AI models, its interpretability needs to be further improved to reach the level of physical estimation models for hydrological forecasting.

VII. CONCLUSION

This research investigated the application of the Bayesian DL model in mountain flood forecasting. The main conclusions are as follows.

(1) We improved the uncertainty forecasting theory for mountain flood mechanism, studied the transfer of spatiotemporal factors caused by small watershed floods to disaster probability, extracted the theoretical influence relationship between disaster factors and disaster results, and established Bayesian DL prediction theoretical method for small watershed mountain flood disasters.

(2) A multi-source and multi-dimensional uncertainty forecasting factor effective information extraction technique has been proposed. By real-time evaluation of multi-source and multi-dimensional disaster causing factors, adopting feature dynamic selection and time-frequency fusion, the effectiveness of disaster information has been improved, the reliability of Bayesian DL prediction models has been enhanced, and effective disaster information has been provided for predicting small watershed mountain floods with missing data and low prediction density.

(3) The models have long-term probability prediction ability, but Bayesian LSTM is superior to Bayesian Linear in terms of confidence interval width, sharpness, and computational resource consumption.

In future work, we plan to challenge the forecasting of higher frequency of water level changes, denser flood periods, and longer prediction times through experiments. In addition, we will further study the reasonable points of reliability and accuracy to obtain higher performance in predicting mountain floods in the watershed.

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